Swiss Re Health Claims Forum

Cardiovascular Risk and Prevention: Is it the Future Yet?

Dr. Abhishek Joshi, PhD, MRCP (Lond.)
Consultant Cardiologist: Bart's Health NHS Trust
Specialist Medical Officer: Swiss Re
Honorary Lecturer: UCL

Objectives

- Understand the CVD NHS prediction and prevention landscape.
- Introduction to the role of genomics.
- Introduction to the role of biomarkers.
- Learn about novel “AI” approaches to risk prediction.

The NHS Long-Term Plan: 2019

Where Were We Going?

NHS Long Term Plan: 2019

- Atrial Fibrillation
- Blood Pressure
- Cholesterol
- Diagnosis of Heart Failure & Valve Disease

Black Swan, or Risk Perennial?

Black Swan, or Risk Perennial?

'I declare COVID over': World Health Organization boss says pandemic is no longer a global health emergency

- The global health leaders said the virus no longer constitutes an emergency
- The UN organization fell short of declaring the pandemic over all-together
- READ MORE: Expert warns that cryptic Covid strains linger across country

Return CVD priorities

Same Goals

Fewer Resources

Greater demand

Narrower bandwidth

Political & economic uncertainty

"40 new hospitals"

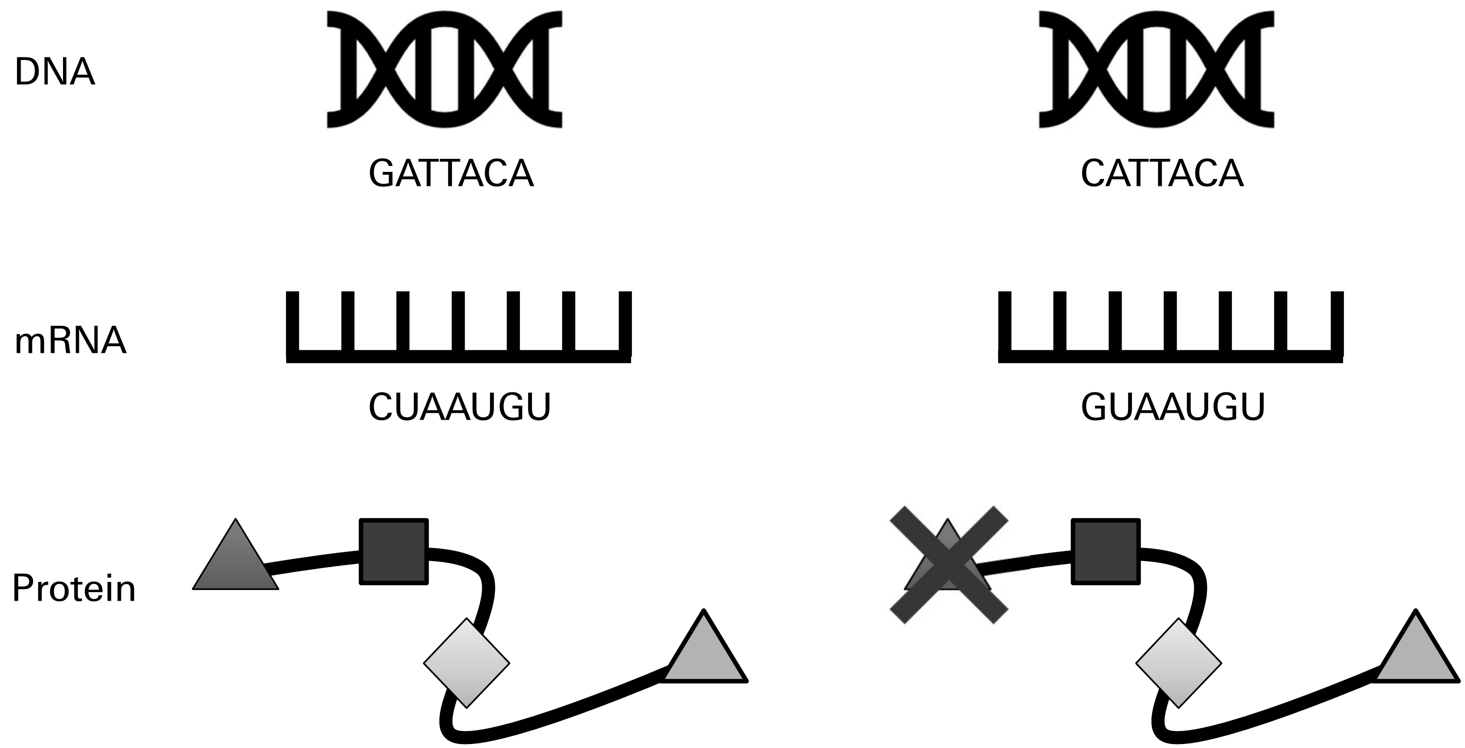
Community diagnostic hubs

Swiss Re

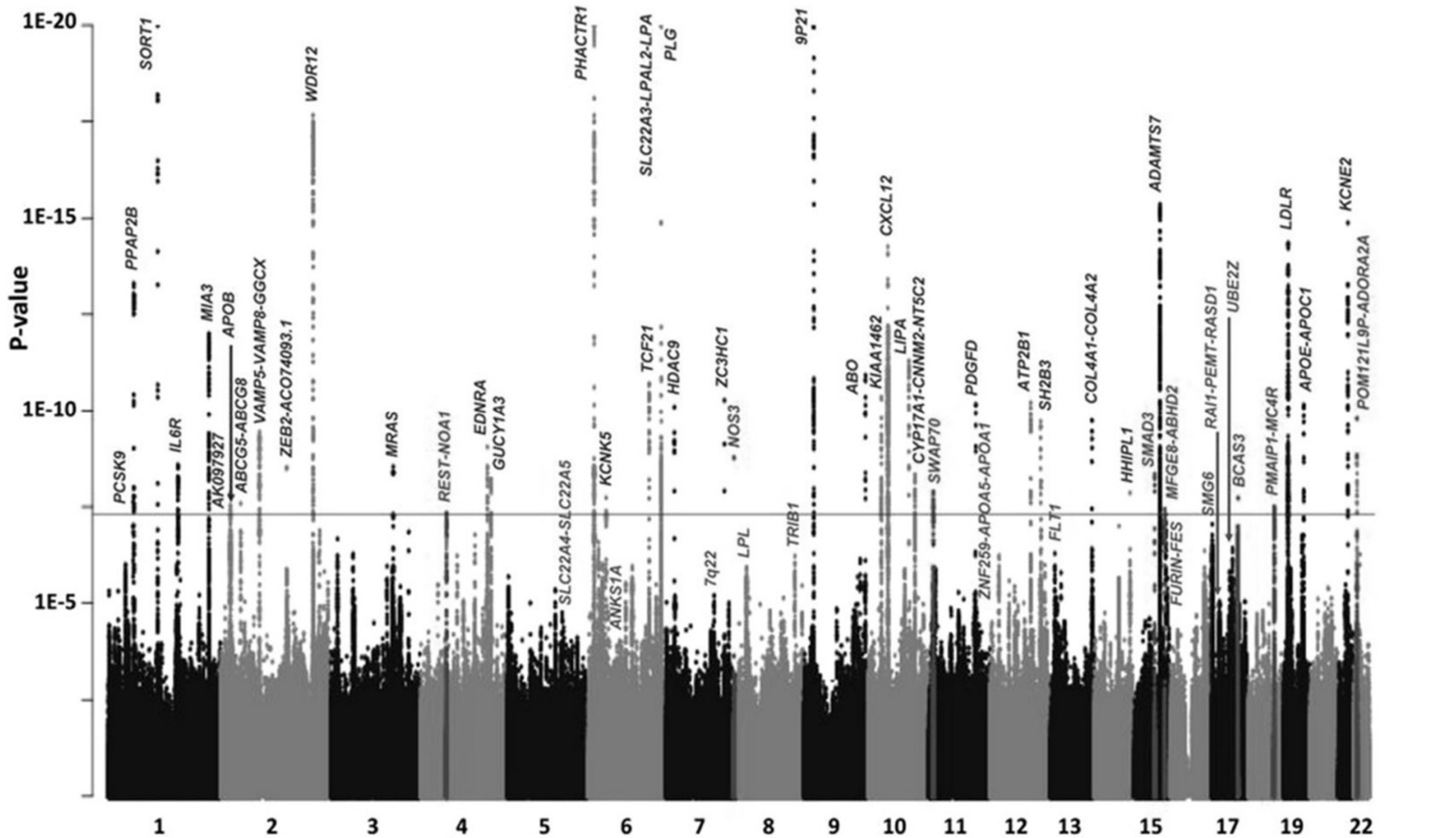
The Promise of Genomics

"Why can't you tell me which day my heart attack will be?"

How does a gene cause a disease?



Genome-wide association studies



Ruth McPherson, Anne Tybjaerg-Hansen, *Circ Res* 2016

9p21: Fame but no fortune

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 AUGUST 2, 2007 VOL. 357 NO. 5

Genomewide Association Analysis of Coronary Artery Disease

Nilesh J. Samani, F.Med.Sci., Jeanette Erdmann, Ph.D., Alistair S. Hall, F.R.C.P., Christian Hengstenberg, M.D., Massimo Mangino, Ph.D., Bjoern Mayer, M.D., Richard J. Dixon, Ph.D., Thomas Meitinger, M.D., Peter Braund, M.Sc., H.-Erich Wichmann, M.D., Jennifer H. Barrett, Ph.D., Inke R. König, Ph.D., Suzanne E. Stevens, M.Sc., Silke Szymczak, M.Sc., David-Alexandre Tregouet, Ph.D., Mark M. Iles, Ph.D., Friedrich Pahlke, M.Sc., Helen Pollard, M.Sc., Wolfgang Lieb, M.D., Francois Cambien, M.D., Marcus Fischer, M.D., Willem Ouwehand, F.R.C.Path., Stefan Blankenberg, M.D., Anthony J. Balmforth, Ph.D., Andrea Baessler, M.D., Stephen G. Ball, F.R.C.P., Tim M. Strom, M.D., Ingrid Brænne, M.Sc., Christian Gieger, Ph.D., Panos Deloukas, Ph.D., Martin D. Tobin, M.F.P.H.M., Andreas Ziegler, Ph.D., John R. Thompson, Ph.D., and Heribert Schunkert, M.D., for the WTCCC and the Cardiogenics Consortium*

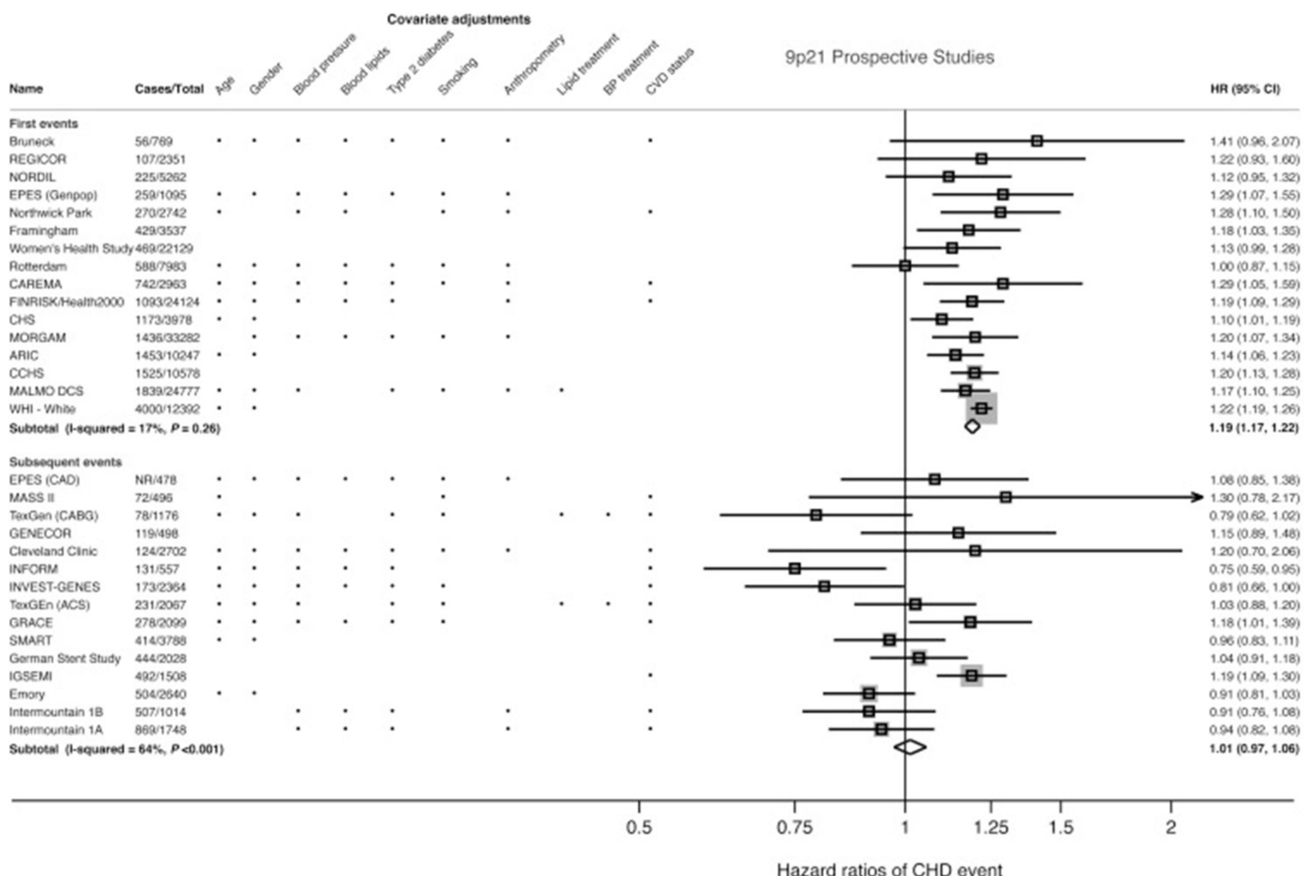
Published: 08 February 2009

Genome-wide association of early-onset myocardial infarction with single nucleotide polymorphisms and copy number variants

Myocardial Infarction Genetics Consortium

Nature Genetics **41**, 334–341 (2009) | Cite this article

9p21: Fame but no fortune



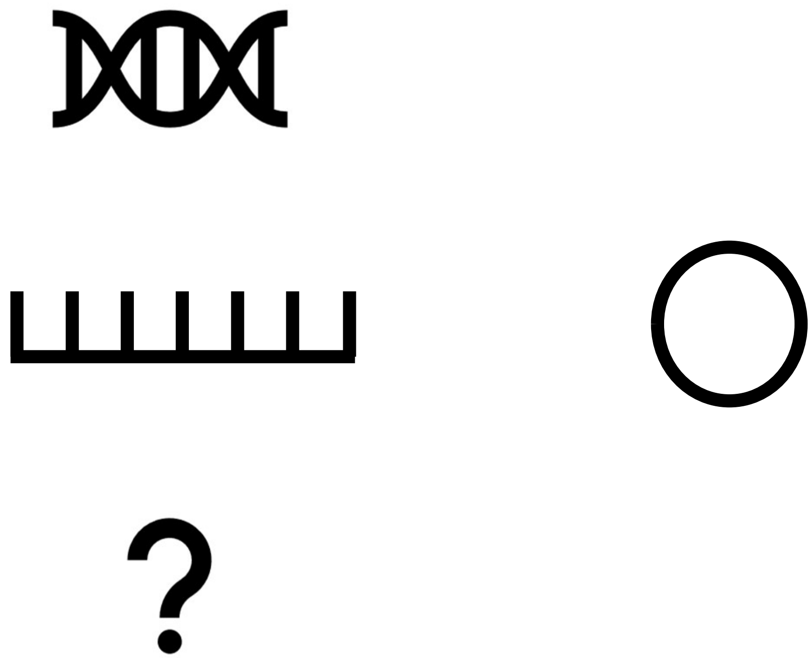
Patel, R. et al, *JACC* 2014

9p21: Molecular mystery

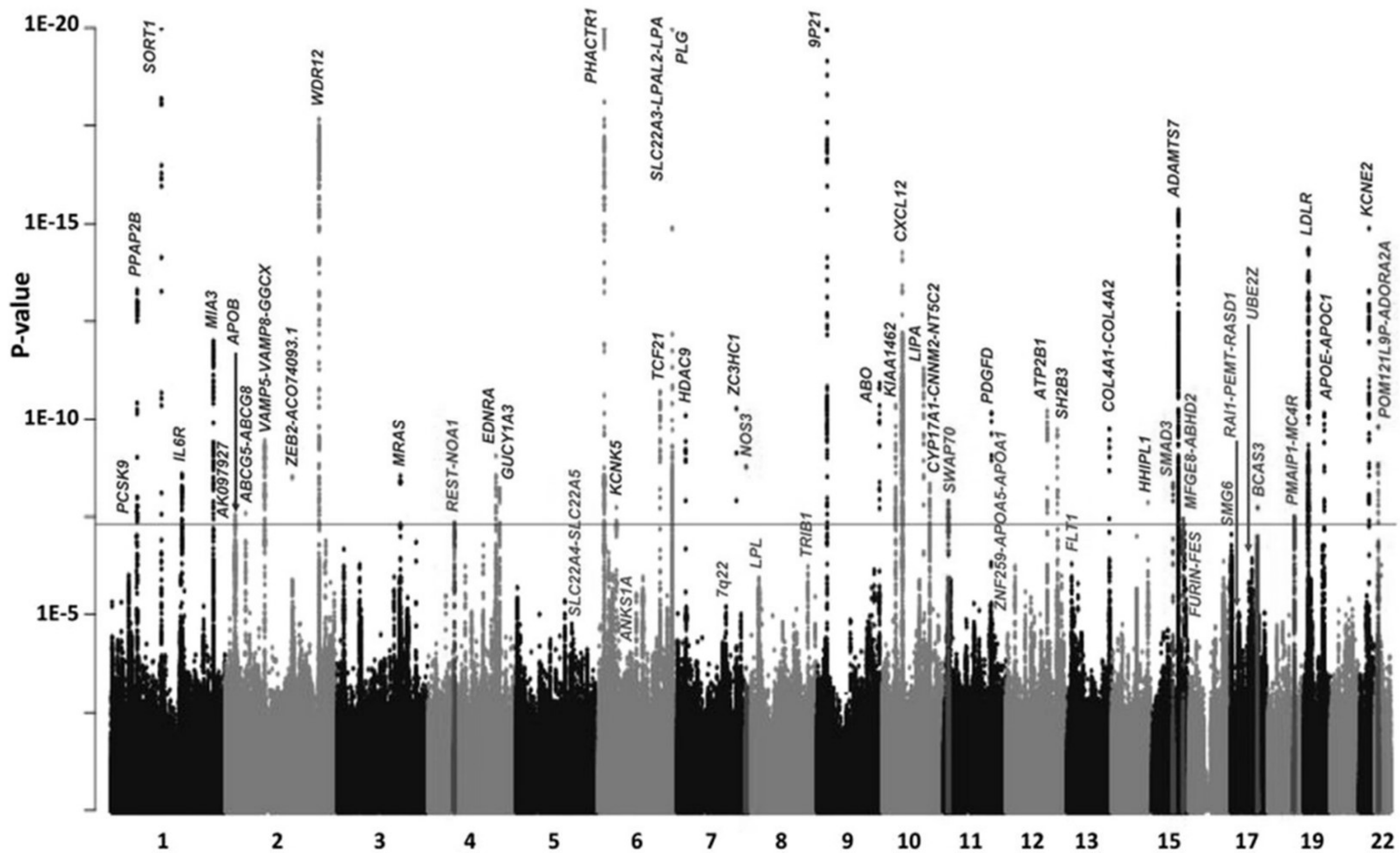
DNA

mRNA

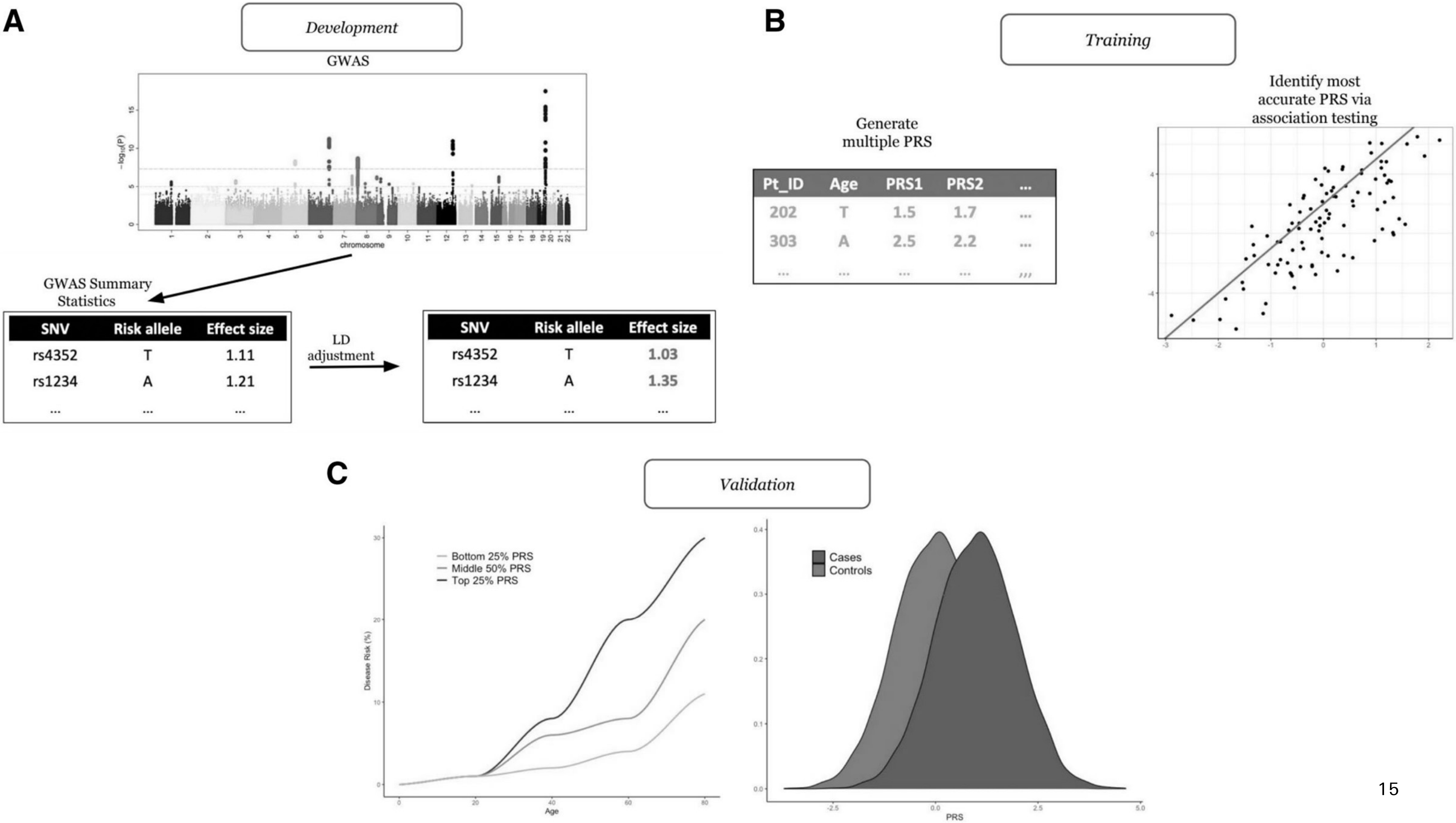
Protein



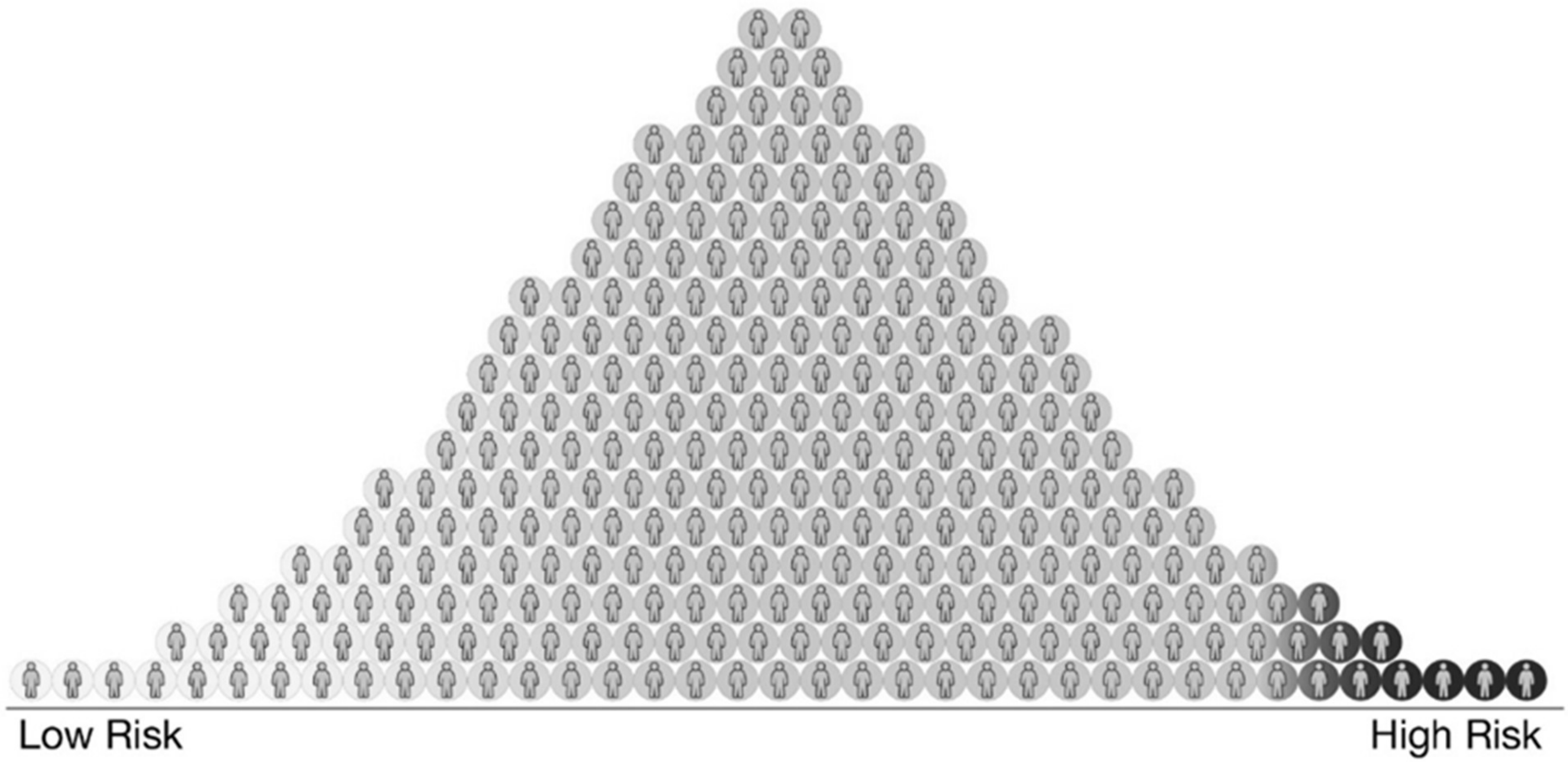
Polygenic Risk Scores: The Emperor's new clothes?



Ruth McPherson, Anne Tybjaerg-Hansen


A
Development
GWAS
chromosome
GWAS Summary Statistics
SNV Risk allele Effect size
rs4352 T 1.11
rs1234 A 1.21
LD adjustment
SNV Risk allele Effect size
rs4352 T 1.03
rs1234 A 1.35
B
Training
Generate multiple PRS
Pt_ID Age PRS1 PRS2 ...
202 T 1.5 1.7 ...
303 A 2.5 2.2 ...
Identify most accurate PRS via association testing
C
Validation
Bottom 25% PRS
Middle 50% PRS
Top 25% PRS
Disease Risk (%)
Age
Cases
Controls
PRS

Polygenic risk scores



JAMA Network

From: **Predictive Accuracy of a Polygenic Risk Score–Enhanced Prediction Model vs a Clinical Risk Score for Coronary Artery Disease**

JAMA. 2020;323(7):636-645. doi:10.1001/jama.2019.22241

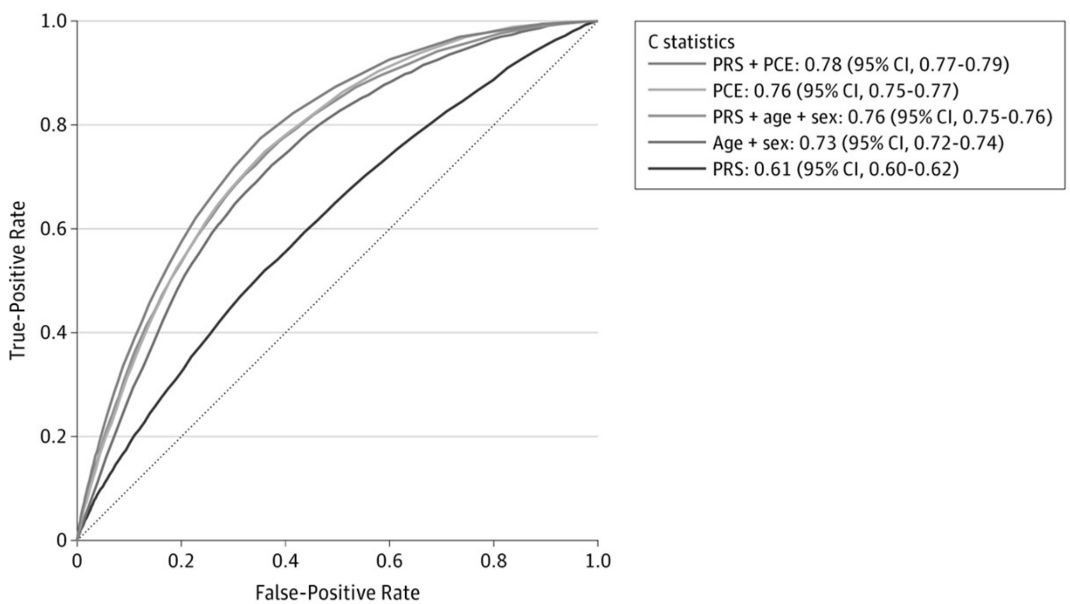
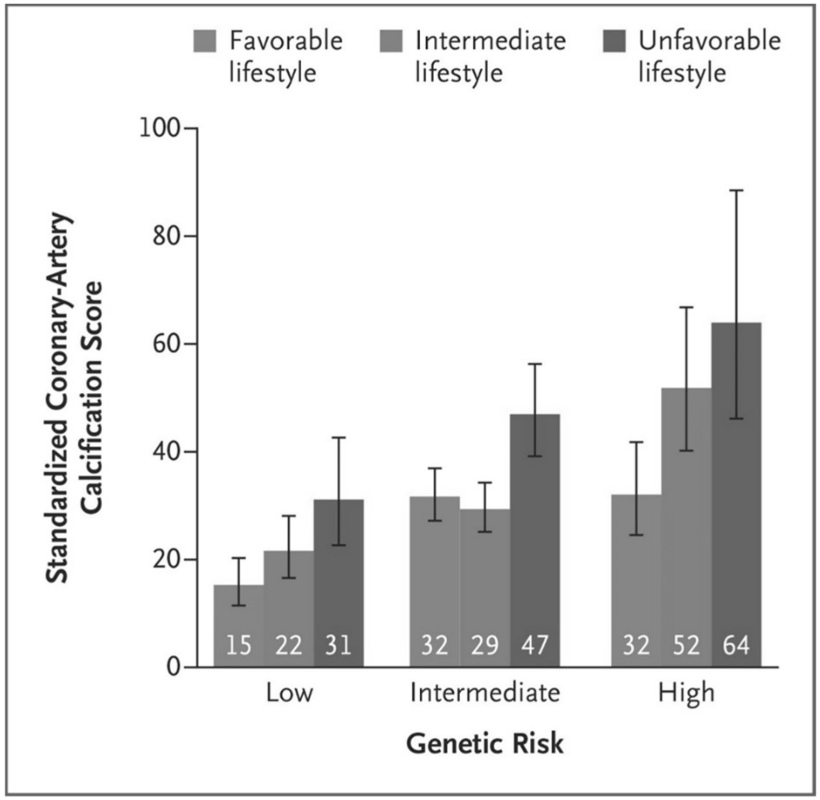


Figure Legend:

Receiver Operator Characteristic Curves and C Statistics for Different Models in Cohort Analyses of 352 660 Participants Aged 40 to 69 Years Old Over a Mean of 8 Years of Follow-up With 6272 Incident Coronary Artery Disease (CAD) EventsSee eTable 2 in the Supplement for the definition of CAD. PCE indicates pooled cohort equations; PRS, polygenic risk score.

Nature or nurture in CVD?



Khera et al, NEJM, 2016

The promise of biomarkers

What is a Biomarker?

A measurable biological feature that is a predictor of disease

- Blood pressure
- Cholesterol
- Coronary Calcium
- Age

- Focus on blood tests to provide "better" information.

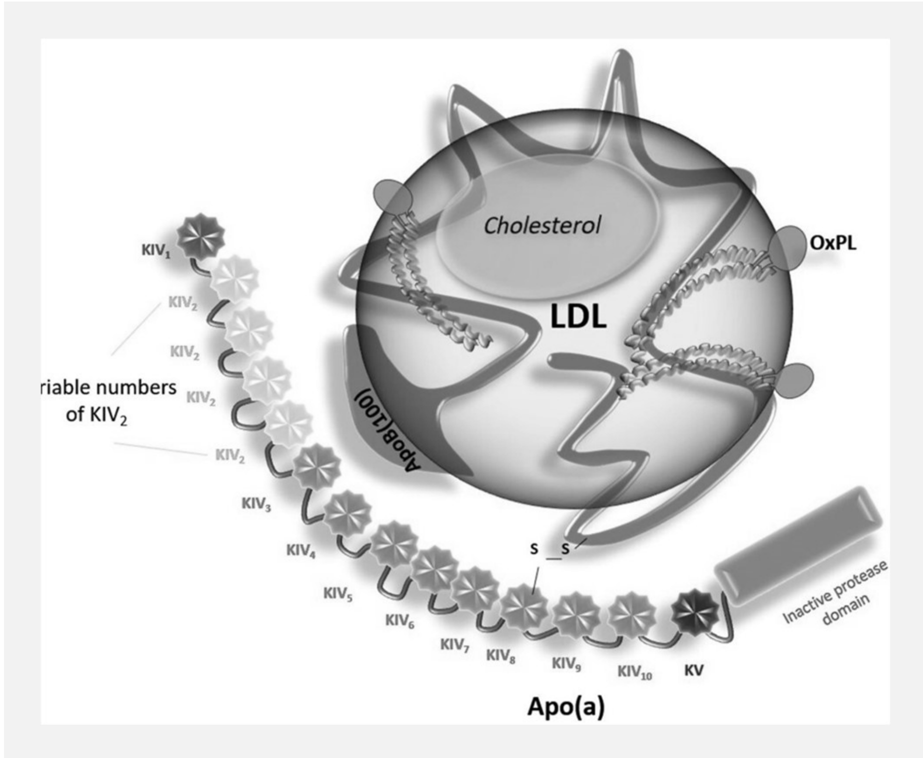
Lp-PLA$_2$

- Identified as a risk factor in 2000s
- Potentially modifiable
- Clinical trials of mAbs failed

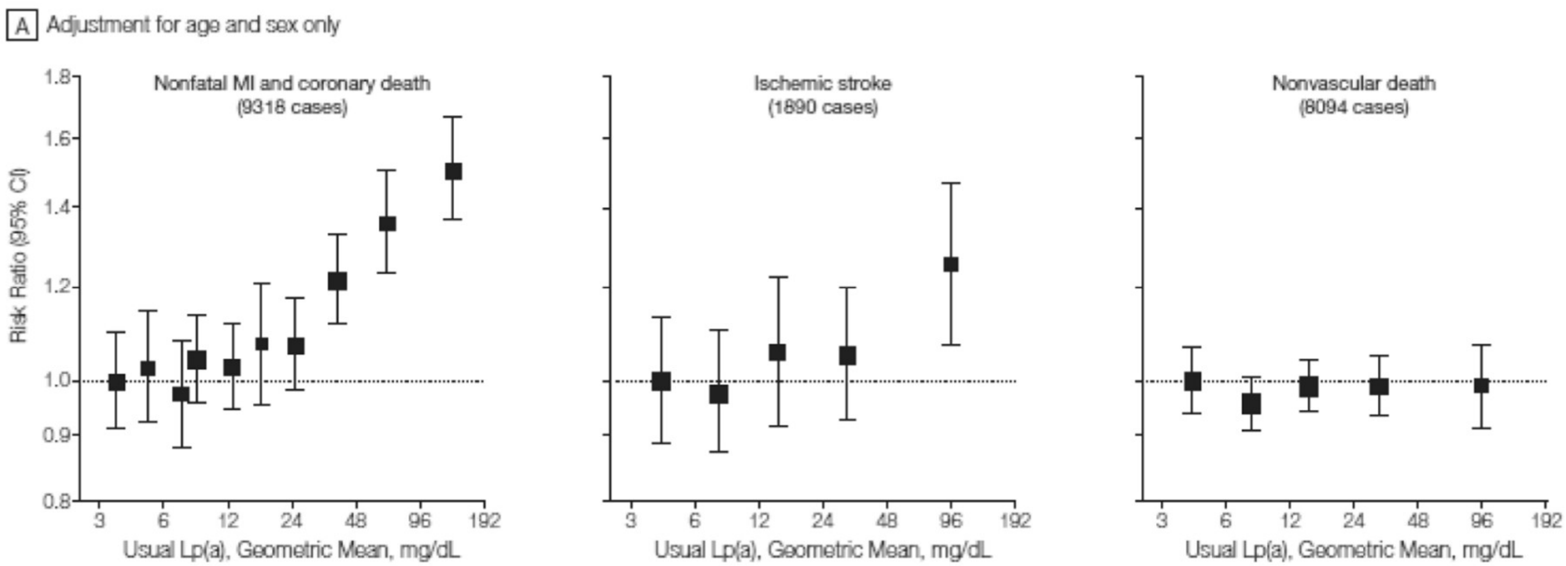
Still measured on insurance panels!

Lipoprotein(a)

- (Another) emerging genetic risk factor



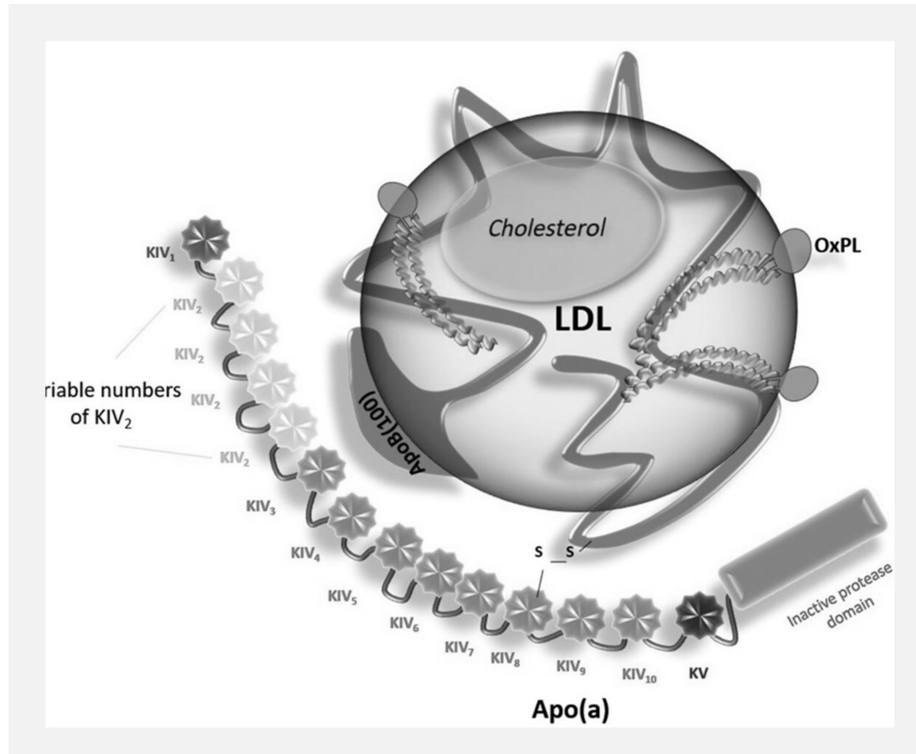
Lipoprotein(a)



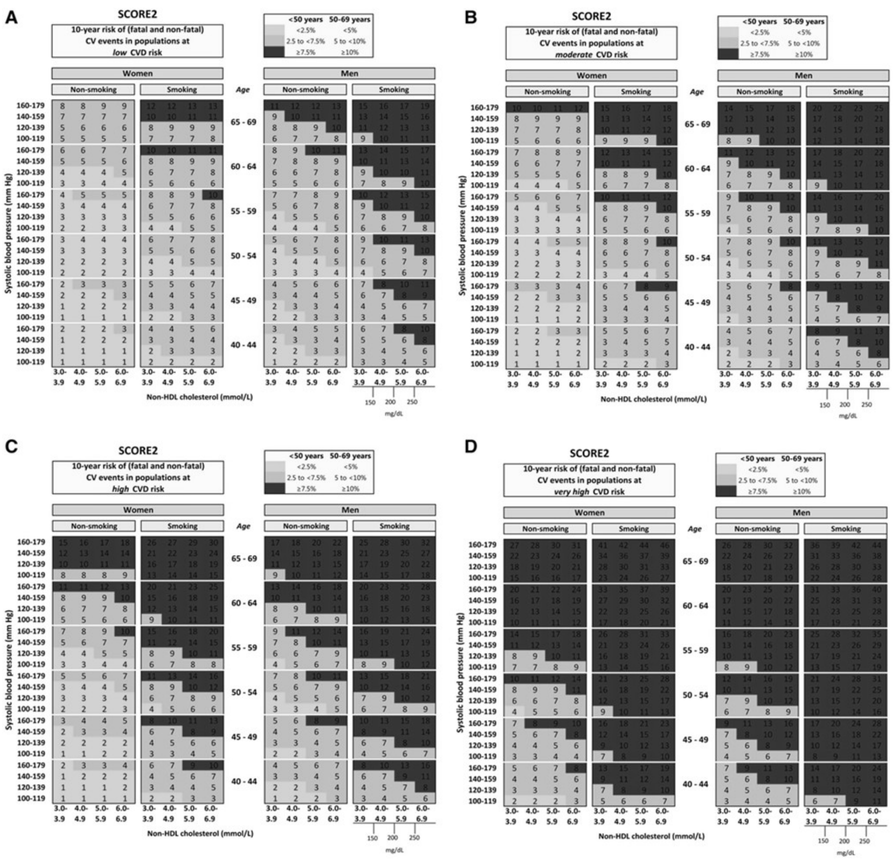
Emerging Risk Factors Collaboration, JAMA 2009

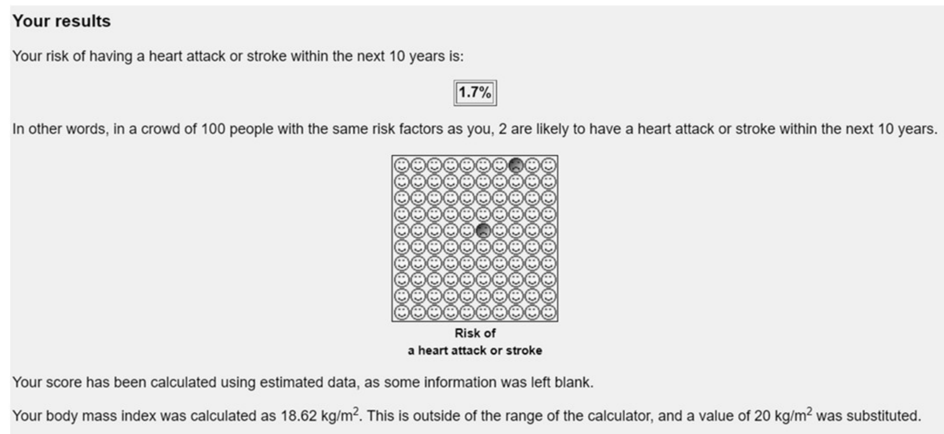
Lipoprotein(a)

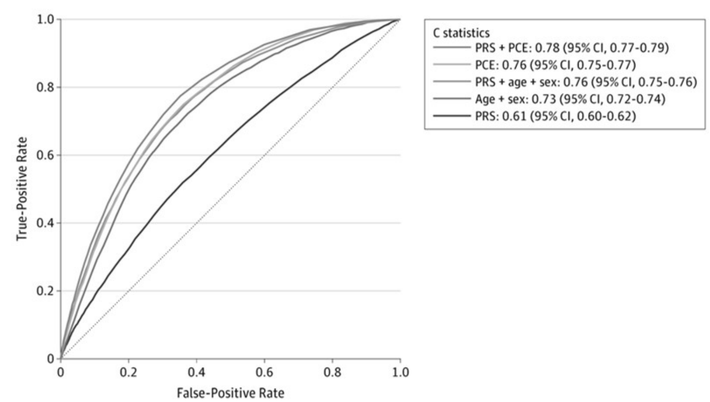
- Potential mRNA therapy in trial
- The next Lp-PLA_2?
- Or the next LDL-C?



Clinical risk scores



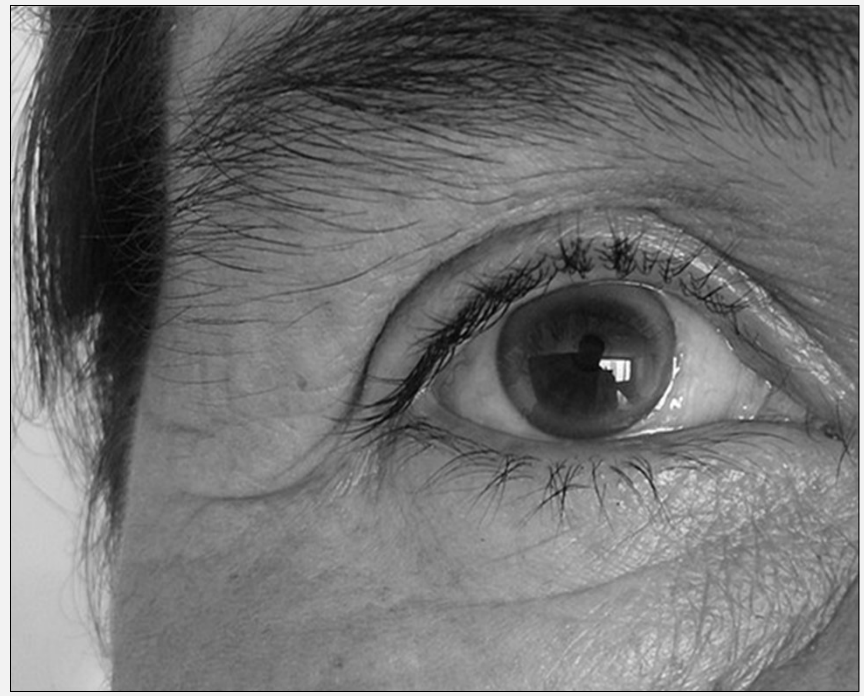




What we already know

...and what should we do about it?

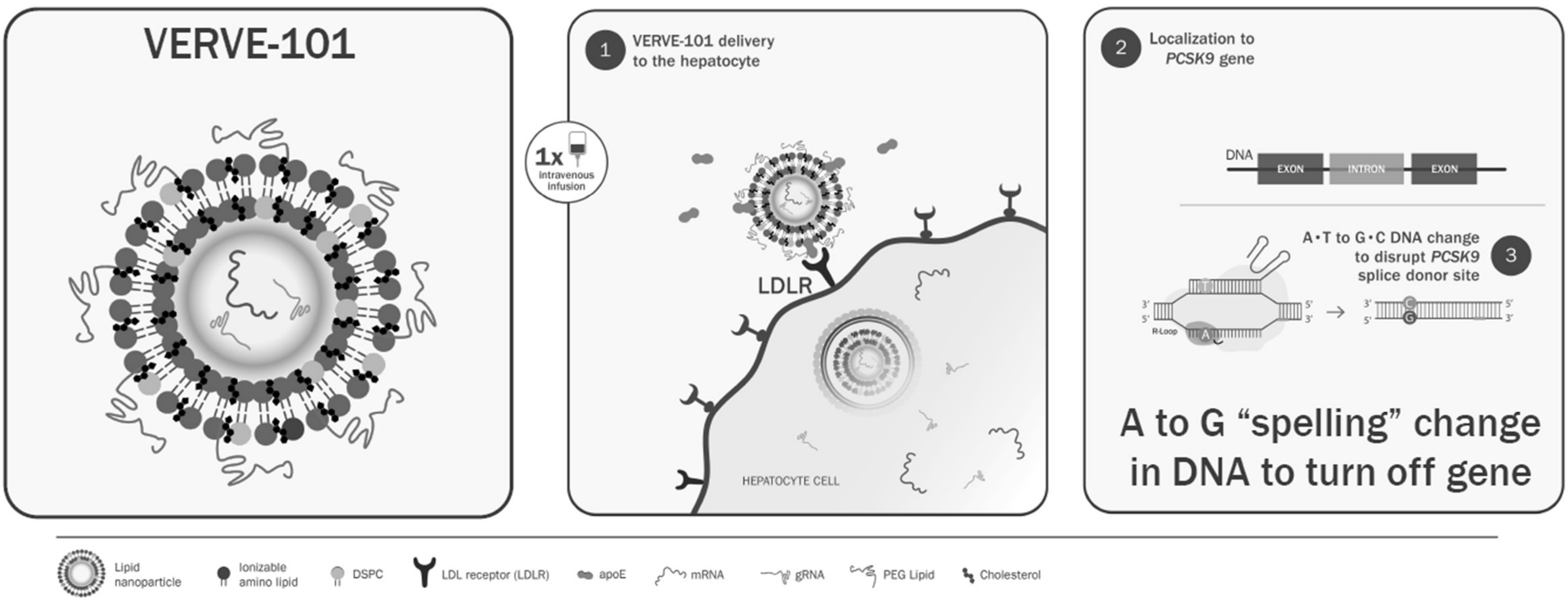
Familial Hypercholesterolaemia



Familial Hypercholesterolaemia

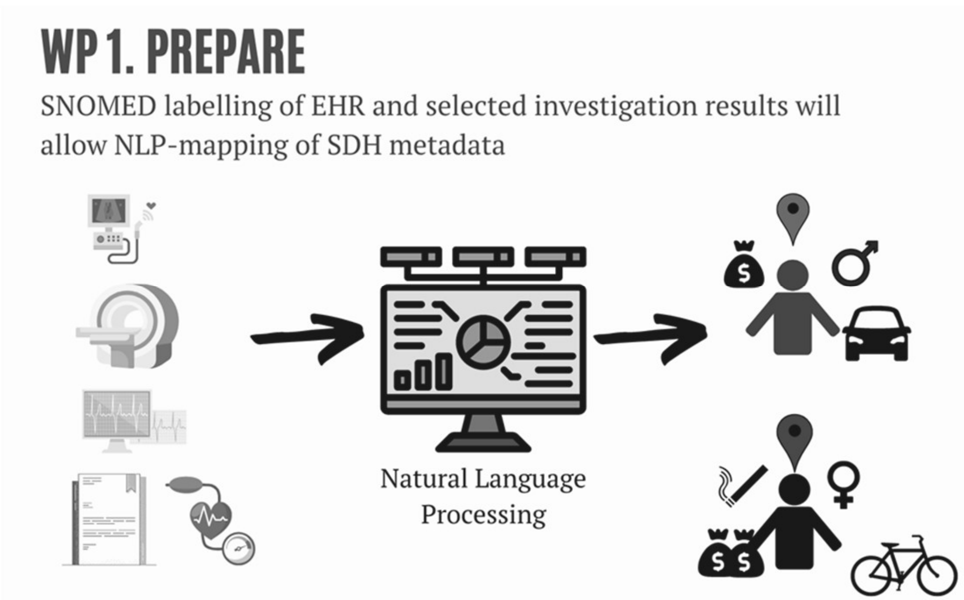
- 1/200 prevalence, genetic cause.
- Extremely premature coronary disease.
- Single gene condition
- Now can be found by national cascade screening programme
- Lives will be saved by treating LDL-C

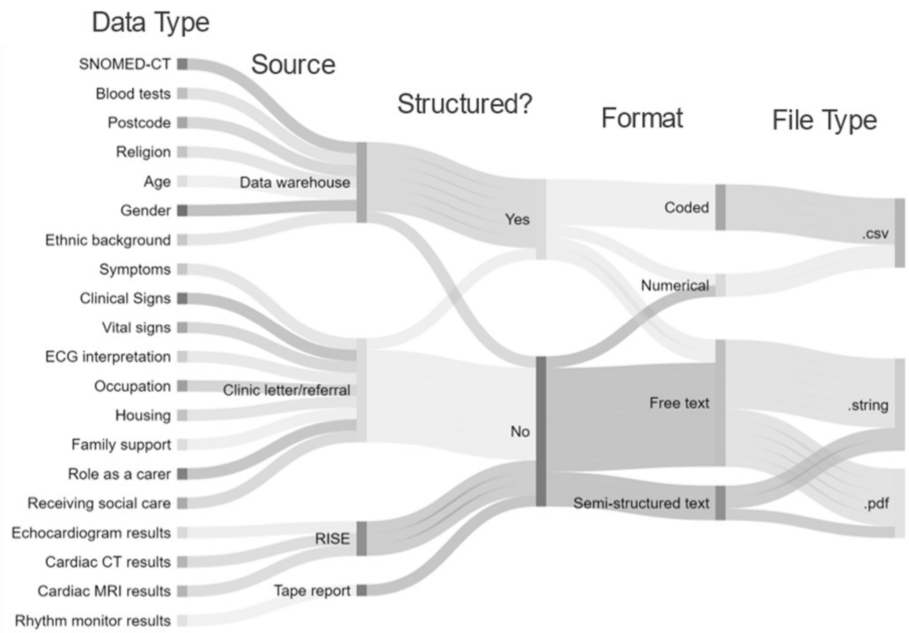
Do We Need More Therapies?



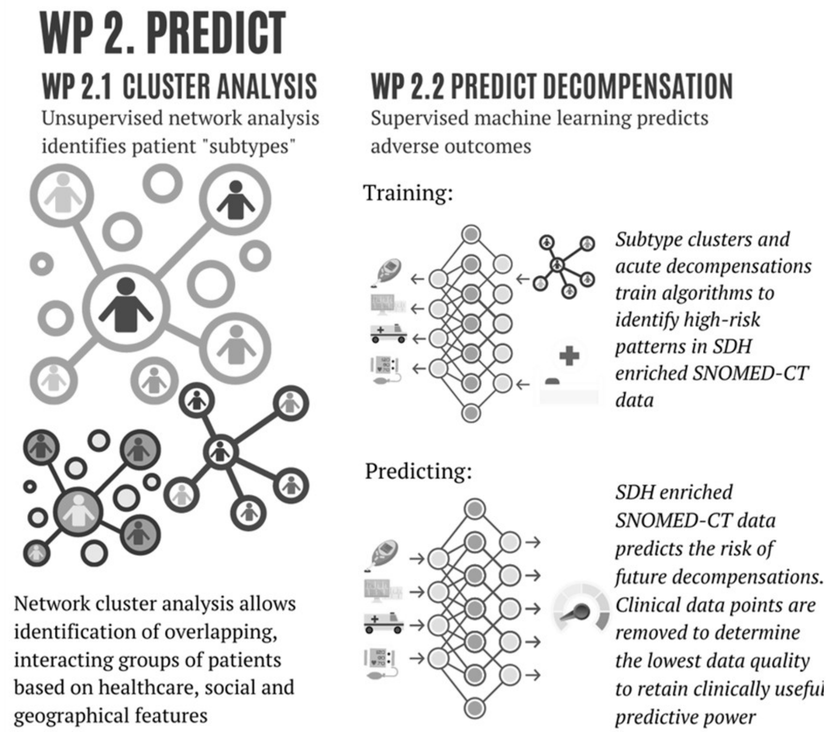
Prediction in the AI era

Natural language processing of EHR

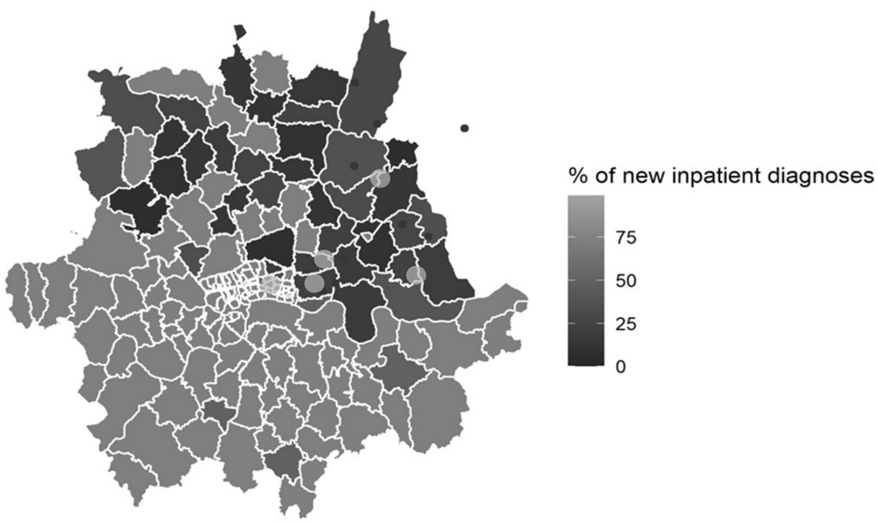


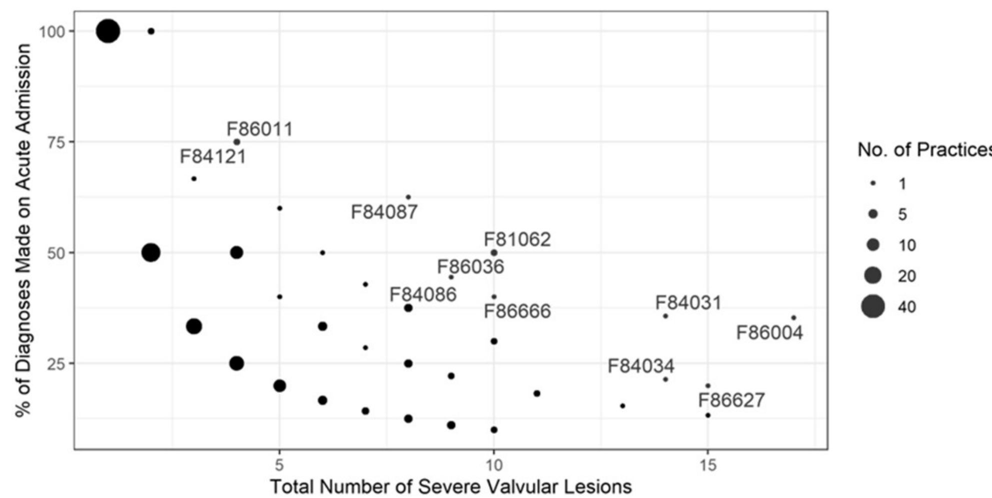


Multi-parameter risk assessment

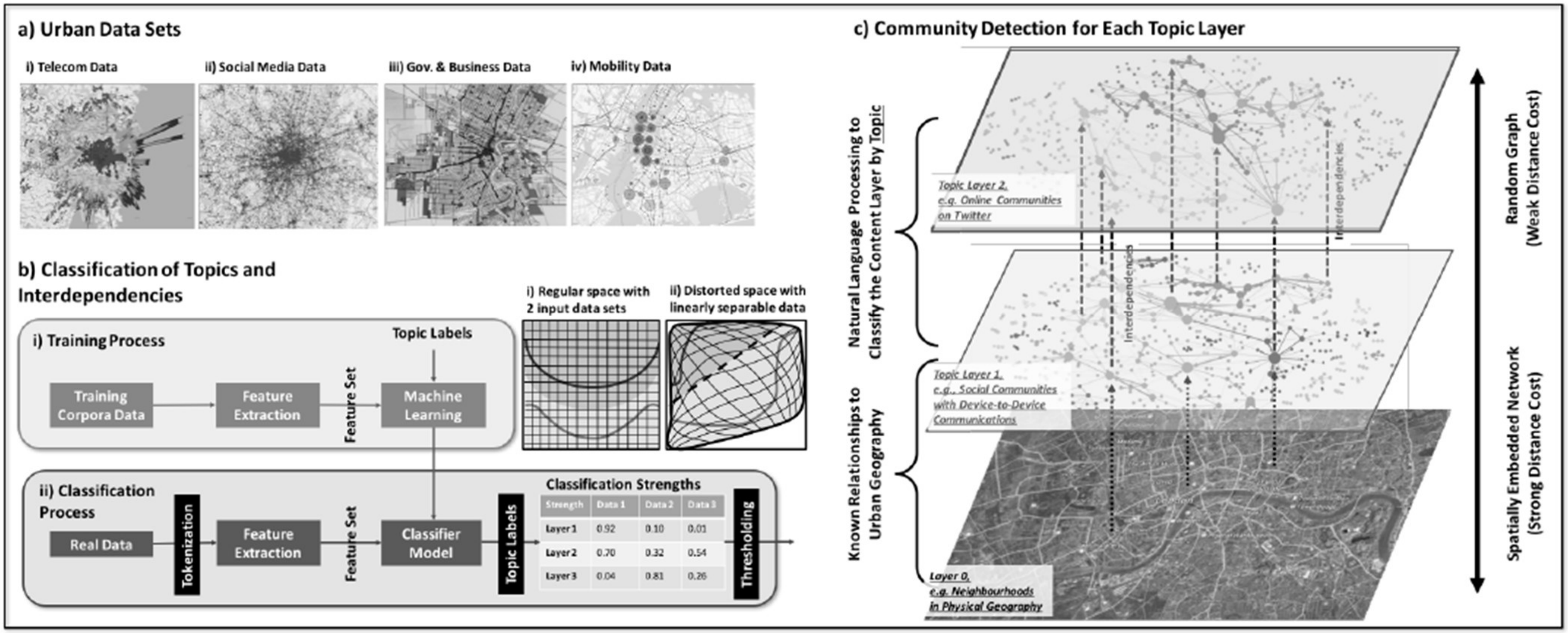


AI approaches to social geography



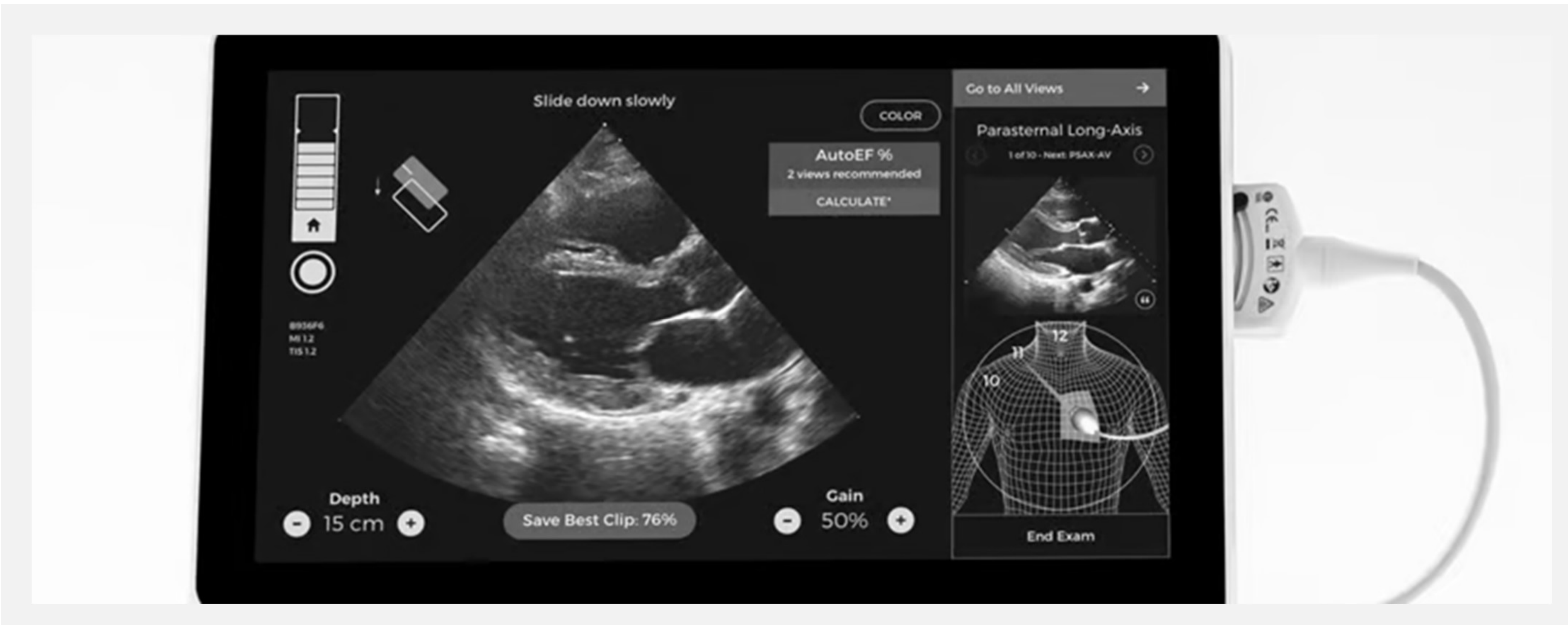


AI approaches to social geography



Strength	Data 1	Data 2	Data 3
Layer 1	0.92	0.10	0.01
Layer 2	0.70	0.32	0.54
Layer 3	0.04	0.81	0.26

AI guided cardiac testing



Bart's PREDICT Schematic

- We will meet the unmet need of undiagnosed valve disease through a targeted early diagnostics approach.
- We will direct resources from the cardiovascular diagnostic hub (CDC) to better target at-risk individuals.
- Targeting will be based on geo-social mapping of risk NE London using established methods.
- Risk features will be extracted from artificial intelligence approaches to the electronic healthcare record.
- Results from our targeting will be fed back into our mapping, improving its accuracy.

Translate EHR to SNOMED-CT

- Team: Life Sciences & DERI
- People: Band 8A + 8C
- Technique: CliniThink
- Cost: £175,000

Draw the Map of NEL Valve Disease

- Team: UCL & Turing Institute
- People: Post-doc
- Technique: Geosocial analysis
- Cost: £125,000

Mobile Diagnostics to Increase Yield

- Team: Diagnostic Hub
- People: Physiologist Band 7
- Technique: AI assisted echo
- Cost: £145,000

- Active learning framework
- Health economic analysis
- Cost: Free

Swiss Re

“Why can´t you tell me when my heart attack will be?”

Summary

Risk Prediction in CVD

NHS Long Term Plan *should* improve CVD rates

We already know everything we need to know.

Hard to beat clinical/traditional measurements

Genomics is (currently) broadly disappointing

New "Personalised" biomarkers are weak predictors

The future is properly implementing the discoveries of the past.

Any questions?

Legal notice

Making a career…ish

Health Claims Forum, February 2024

Willie Hamilton, MD BSc, FRCP, FRCGP

Chief Medical Officer, The Exeter and
Professor of Primary Care Diagnostics, University of Exeter

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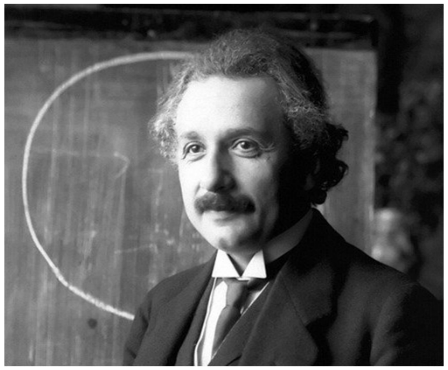
Let's get philosophical…

"If a tree were to fall on an island where there were no human beings would there be any sound?"
Bishop Berkeley [attributed]



"The moon does not exist if nobody is looking at it."
Albert Einstein

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However good your research is if it’s not implemented, it’s as much use as a…

….handbrake on a canoe,
vote in a dictatorship,
loudhailer to a deaf mute,
grief at a wedding,

condom in a monastery.
inflatable dartboard,
spoon in a knife-fight,
screen door on a submarine…

Paul Steven Laurence

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Cancer diagnosis in the NHS and the insurance world: getting research into practice

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Did you know the UK comes bottom of most cancer league tables?

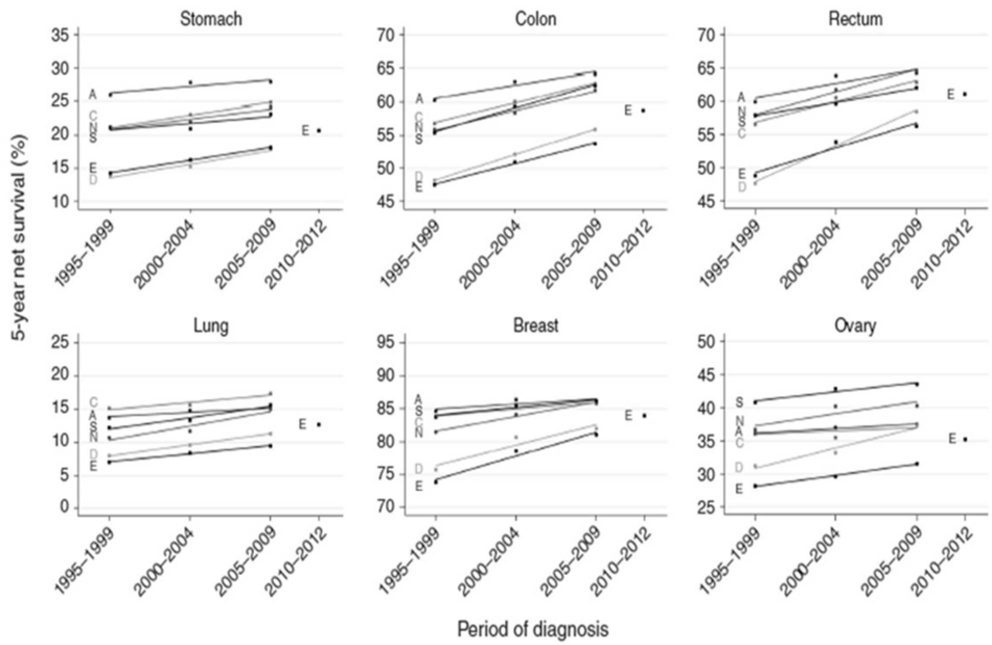

Figure 2. Trends in 1- and 5-year net survival in Australia (A), Canada (C), Denmark (D), England (E), Norway (N) and Sweden (S) by period of diagnosis. Estimates of net survival are presented for the calendar periods of diagnosis 1995–1999, 2000–2004 and 2005–2009. Simple linear regression lines are presented for each combination of country and cancer using data from these three periods, to indicate the average change in survival. An estimate of net survival for England only is also presented for the calendar period of diagnosis 2010–2012.

FULL PAPER

BJC

British Journal of Cancer (2015) 113, 848–860 | doi: 10.1038/bjc.2015.265

Keywords: cancer survival; international comparison; population-based; cancer registries

Is England closing the international gap in cancer survival?

Sarah Walters[*,1], Sara Benitez-Majano[1], Patrick Muller[1], Michel P Coleman[1], Claudia Allemani[1], John Butler[1,2], Mick Peake[3], Marianne Grønlie Guren[4,5], Bengt Glimelius[6], Stefan Bergström[7], Lars Påhlman[8] and Bernard Rachet[1]

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England may beat Croatia at football, but…

- Patients in Denmark, UK and Eastern European countries have lower survival than patients in other neighbouring countries.
- A large amount of the survival differences relates to the first year after diagnosis, probably reflecting the higher proportion of cases diagnosed at an advanced stage.
- So, do UK patients have delayed diagnoses? And if so, does this mean some have cancer symptoms when they apply for insurance?


European Journal of Cancer (2015) 51, 2120–2129

Available at www.sciencedirect.com

ScienceDirect

journal homepage: www.ejcancer.com

ELSEVIER

EJC

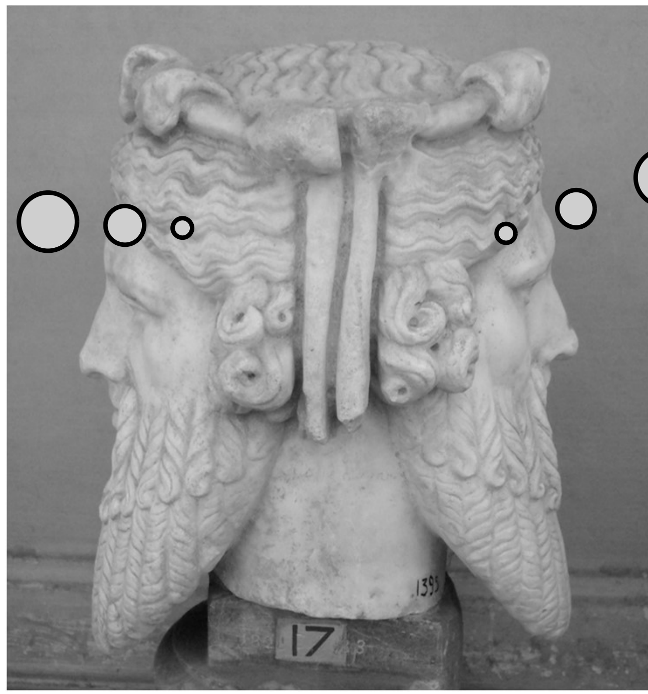
Age and case mix-standardised survival for all cancer patients in Europe 1999–2007: Results of EUROCARE-5, a population-based study

CrossMark

Paolo Baili[a], Francesca Di Salvo[a,*], Rafael Marcos-Gragera[b], Sabine Siesling[c,d], Sandra Mallone[e], Mariano Santaquilani[f], Andrea Micheli[a,g], Roberto Lillini[h,i], Silvia Francisci[e], and the EUROCARE-5 Working Group[1]

Around half of the problem is ascribed to delays in diagnosis.
What if an applicant says she has abdominal pain and diarrhoea?

Abdominal pain and diarrhoea = Irritable Bowel Syndrome

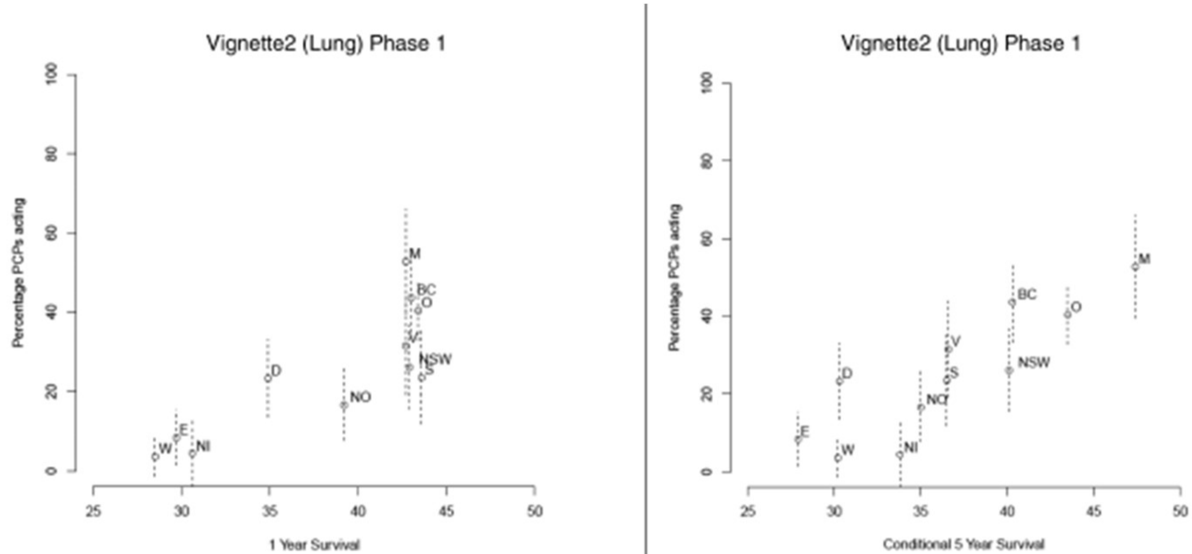


Abdominal pain and diarrhoea = Colorectal cancer

Phase 1: Having spotted there's a clinical problem (or opportunity)

- Describe the problem (or read what others have described)

It may well be GPs are less keen to test



B = 37.544	Std. Error = 5.963	p-value = <0.001	B = 31.916	Std. Error = 5.653	p-value = <0.001

Open Access Research

BMJ Open

Explaining variation in cancer survival between 11 jurisdictions in the International Cancer Benchmarking Partnership: a primary care vignette survey

Peter W Rose,[1] Greg Rubin,[2] Rafael Perera-Salazar,[1] Sigrun Saur Almberg,[3] Andriana Barisic,[4] Martin Dawes,[5] Eva Grunfeld,[6,7] Nigel Hart,[8] Richard D Neal,[9] Marie Pirotta,[10] Jeffrey Sisler,[11] Gerald Konrad,[12] Berit Skjødeberg Toftegaard,[13] Hans Thulesius,[14] Peter Vedsted,[15] Jane Young,[16] Willie Hamilton,[17] The ICBP Module 3 Working Group*

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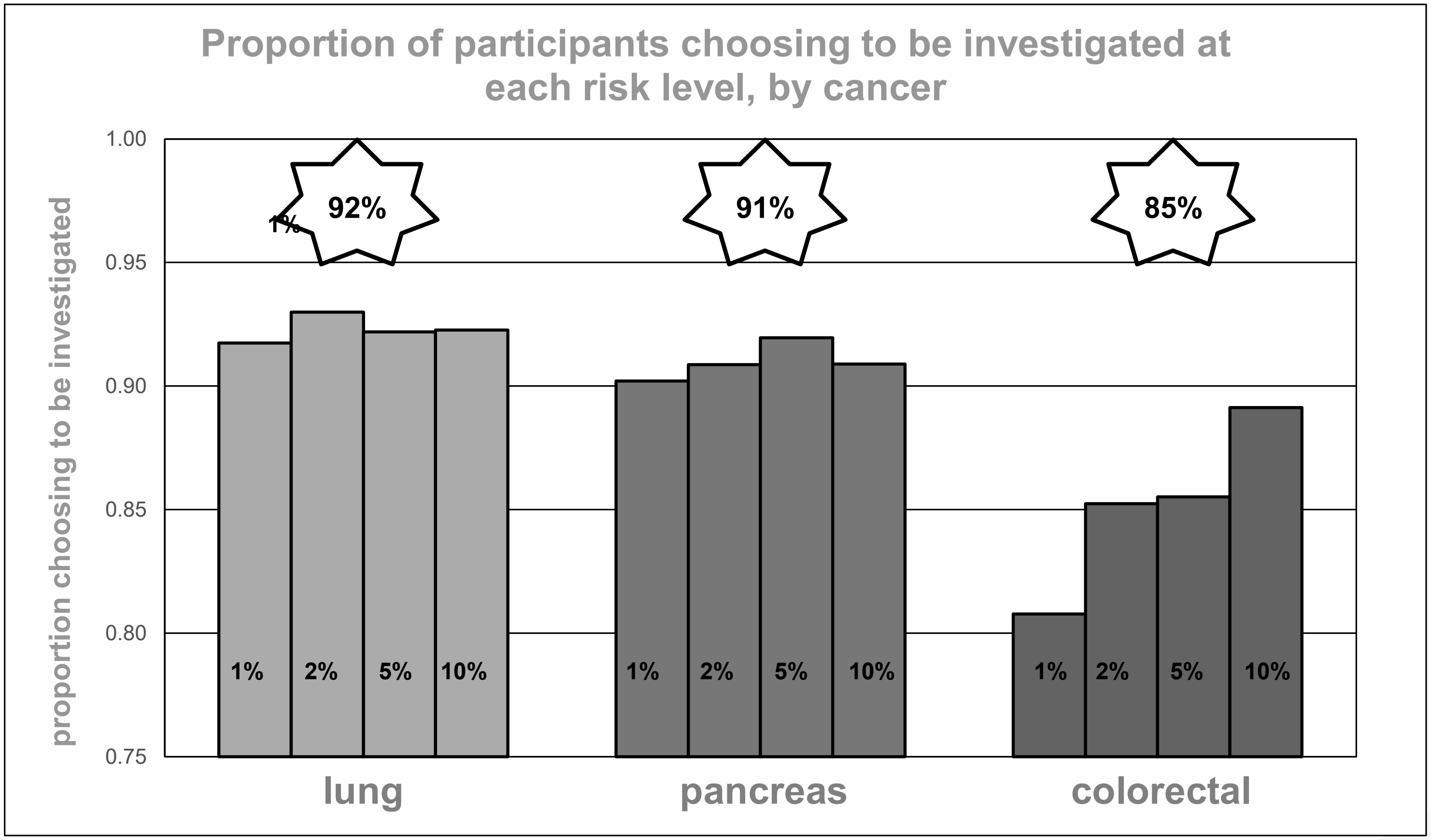
Even though patients want to be tested…



Preferences for cancer investigation: a vignette-based study of primary-care attendees

Jonathan Banks, Sandra Hollinghurst, Lin Bigwood, Tim J Peters, Fiona M Walter, Willie Hamilton

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There’s a gap between what patients want and what was on offer in the NHS

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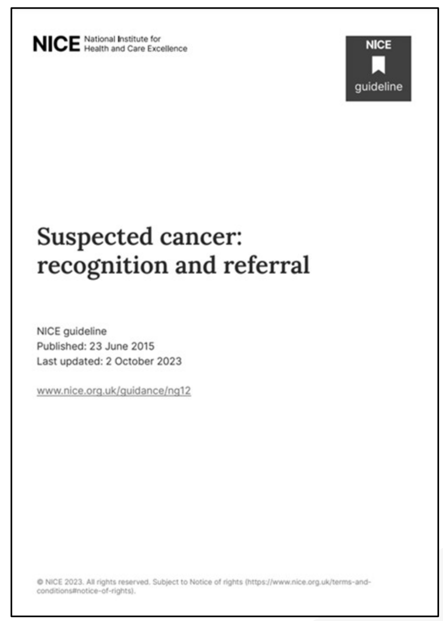
Phase 2: Think of a way of helping

In cancer, the improvements were broadly:

- ► Inform the public of the symptoms of cancer
  [DH/Government funding]
- ► Help GPs to identify patients with cancer
  [Research programme, funded by NIHR, CRUK and others]
- ► Get the results of this research into practice
  [DH/Government funding/NICE/researcher-led initiatives]


NICE National Institute for Health and Care Excellence

NICE guideline

Suspected cancer: recognition and referral

NICE guideline
Published: 23 June 2015
Last updated: 2 October 2023

www.nice.org.uk/guidance/ng12

Phase 3: Identify the symptoms of cancer

- In a near 15-year phase of work where we have identified the key symptoms of cancer for all main cancers, one-by-one
- We have meticulously studied the GP records of 88.150 patients with one of 18 cancer types, plus 246,064 age sex and practice controls without cancer

British Journal of Cancer (2009) 101, S80–S86

www.bjcancer.com

Full Paper

The CAPER studies: five case-control studies aimed at identifying and quantifying the risk of cancer in symptomatic primary care patients

W Hamilton*,[1]
[1] Department of Community Based Medicine, NIHR School for Primary Care Research, University of Bristol, 25-27 Belgrave Road, Bristol BS8 4AA, UK

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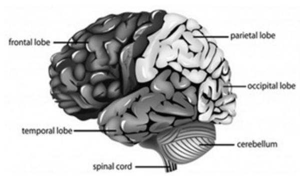
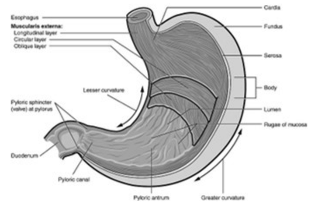
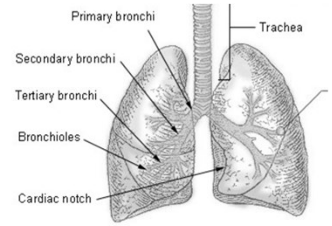
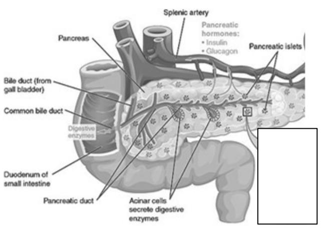
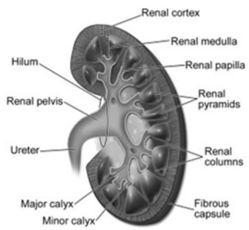
We know the risk profiles for most cancers

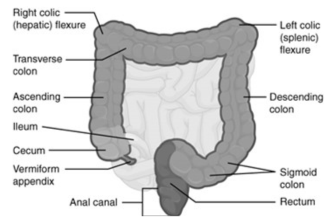


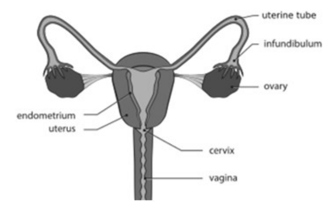


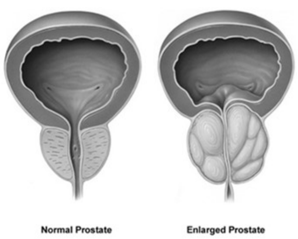


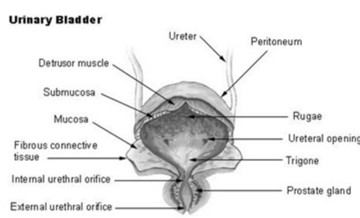


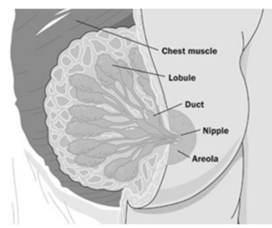












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Phase 4: Present your work in a user-friendly way

Table 2 Symptoms of cancer recorded in primary care, their prevalence in cases (%) and their likelihood ratios (LR)

Colorectal cancer			Prostate cancer			Lung cancer			Brain tumours		
Symptom	**%**	**LR**	**Symptom**	**%**	**LR**	**Symptom**	**%**	**LR**	**Symptom**	**%**	**LR**
Rectal bleeding	42	10	Urinary retention	15	9	Haemoptysis	20	13	New seizure	4	96
Loss of weight	27	5.1	Hesitancy	17	9	Loss of weight	27	6	Motor loss	9	21
Abdominal pain	42	4.5	Impotence	31	9	Loss of appetite	19	5	Confusion	3	16
Diarrhoea	38	3.9	Frequency	47	7	Dyspnoea	56	4	Weakness	3	11
Constipation	26	1.8	Nocturia	29	6	Chest or rib pain	42	3	Headache	10	7
			Haematuria	15	3	Fatigue	35	2	Memory loss	1	3
			Loss of weight	10	2	Cough	65	2	Visual disorder	1	3

Note: The first column shows the percentage of cases with the symptom recorded in their notes before diagnosis, and the second column is the positive likelihood ratio. For simplicity, all figures have been rounded to integers.

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Is this better?

- Using absolute risks…

Constipation	Diarrhoea	Rectal bleeding	Loss of weight	Abdominal pain	Abdominal tenderness	Abnormal rectal exam	Haemoglobin 10–13g dl^{-1}	Haemoglobin < 10g dl^{-1}	
0.42 0.3, 0.5	**0.94** 0.7, 1.1	**2.4** 1.9, 3.2	**1.2** 0.9, 1.6	**1.1** 0.9, 1.3	**1.1** 0.8, 1.5	**1.5** 1.0, 2.2	**0.97** 0.8, 1.3	**2.2** 1.6, 3.1	PPV as a single symptom
0.81 0.5, 1.3	**1.1** 0.6, 1.8	**2.4** 1.4, 4.4	**3.0** 1.7, 5.4	**1.5** 1.0, 2.2	**1.7** 0.9, 3.4	**2.6**	**1.2** 0.6, 2.7	**2.6**	Constipation
	1.5 1.0, 2.2	**3.4** 2.1, 6.0	**3.1** 1.8, 5.5	**1.9** 1.4, 2.7	**2.4** 1.3, 4.8	**11**	**2.2** 1.2, 4.3	**2.9**	Diarrhoea
		6.8	**4.7**	**3.1** 1.9, 5.3	**4.5**	**8.5**	**3.6**	**3.2**	Rectal bleeding
			1.4 0.8, 2.6	**3.4** 2.1, 6.0	**6.4**	**7.4**	**1.3** 0.7, 2.6	**4.7**	Loss of weight
				3.0 1.8, 5.2	**1.4** 0.3, 2.2	**3.3**	**2.2** 1.1, 4.5	**6.9**	Abdominal pain
					1.7 0.8, 3.7	**5.8**	**2.7**	**>10**	Abdominal tenderness

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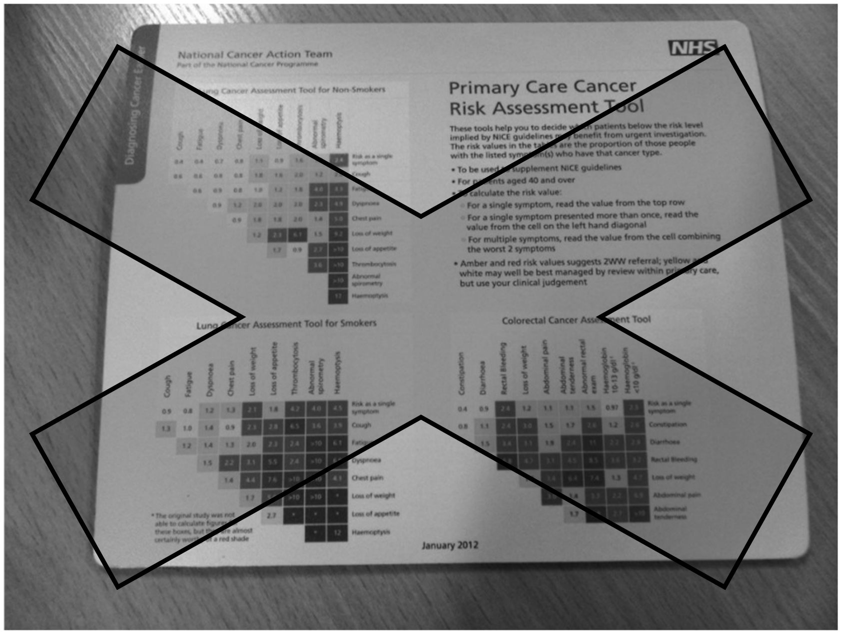
Entering the 21st century

We’ve all seen mouse mats, coffee cups, calendars, but…



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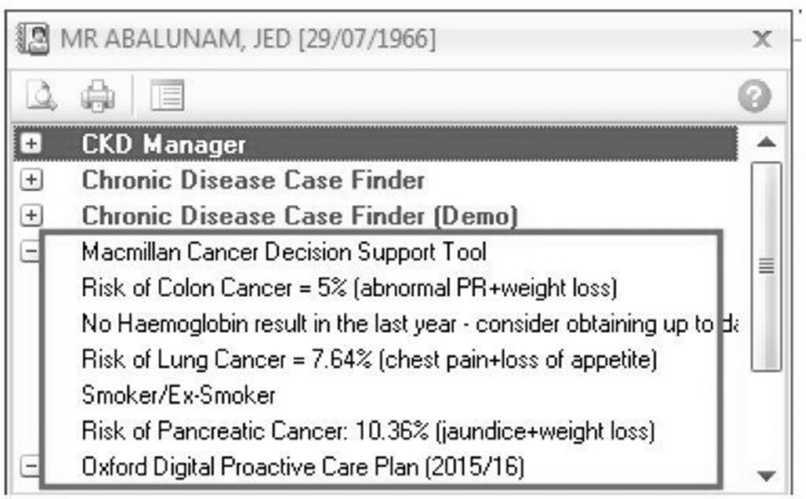
…and entering GP IT systems…

- A prompt can pop up at the start of the consultation, alerting the GP to the possibility of cancer.

Though, prompts are not always popular.



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The ERICA trial

A cluster RCT in 477 English practices

- An electronic prompt appears on the GP's screen if features of cancer add up to a 2% risk of any one cancer
- Main outcome: stage at diagnosis
- Results due 2025…

Phase 5: Get charity and philanthropic support for your work

MACMILLAN CANCER SUPPORT

Target Ovarian Cancer

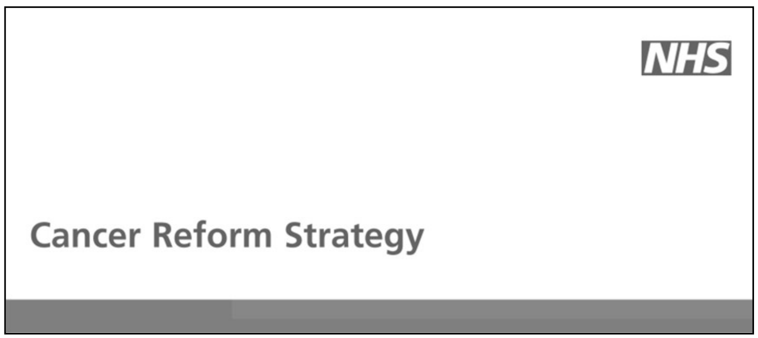
CANCER RESEARCH UK

Pancreatic Cancer UK

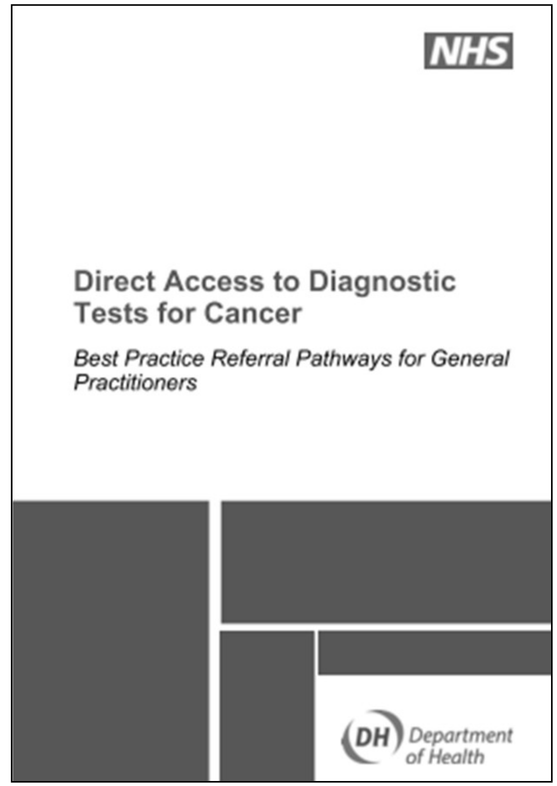
PROSTATE CANCER UK

Bowel Cancer UK

Phase 6: Get political support for your work


NHS

Cancer Reform Strategy


NHS

Direct Access to Diagnostic Tests for Cancer

Best Practice Referral Pathways for General Practitioners

DH Department of Health

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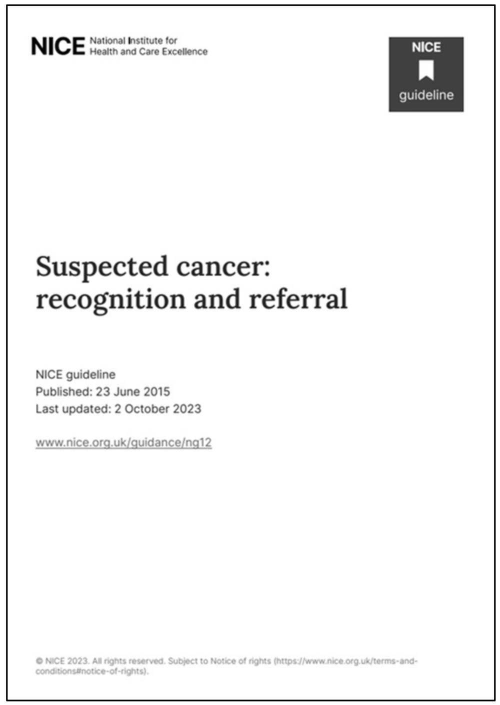
Phase 7: Get NICE to use your work

- This may involve considerable self-sacrifice (honest)
- And recovery may be slow… but



NICE National Institute for Health and Care Excellence

NICE guideline

Suspected cancer: recognition and referral

NICE guideline
Published: 23 June 2015
Last updated: 2 October 2023

www.nice.org.uk/guidance/ng12

- How else can you influence £1bn of annual NHS spending?

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Phase 8: Sit back and await results…

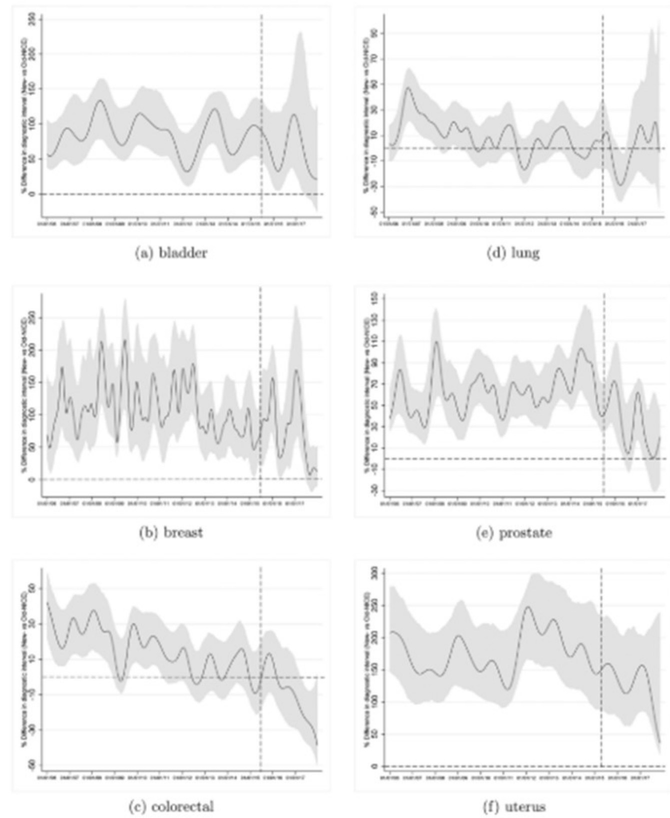
And GPs are more adaptable than you may think

CanTest
Right place, right time, by your family doctor

Contents lists available at ScienceDirect
Cancer Epidemiology
journal homepage: www.elsevier.com/locate/canep
ELSEVIER

Trends in time to cancer diagnosis around the period of changing national guidance on referral of symptomatic patients: A serial cross-sectional study using UK electronic healthcare records from 2006–17

Sarah Price[a,*], Anne Spencer[b], Xiaohui Zhang[c], Susan Ball[d], Georgios Lyratzopoulos[e], Ruben Mujica-Mota[f], Sal Stapley[g], Obioha C Ukoumunne[h], Willie Hamilton[i]



CANCER RESEARCH UK

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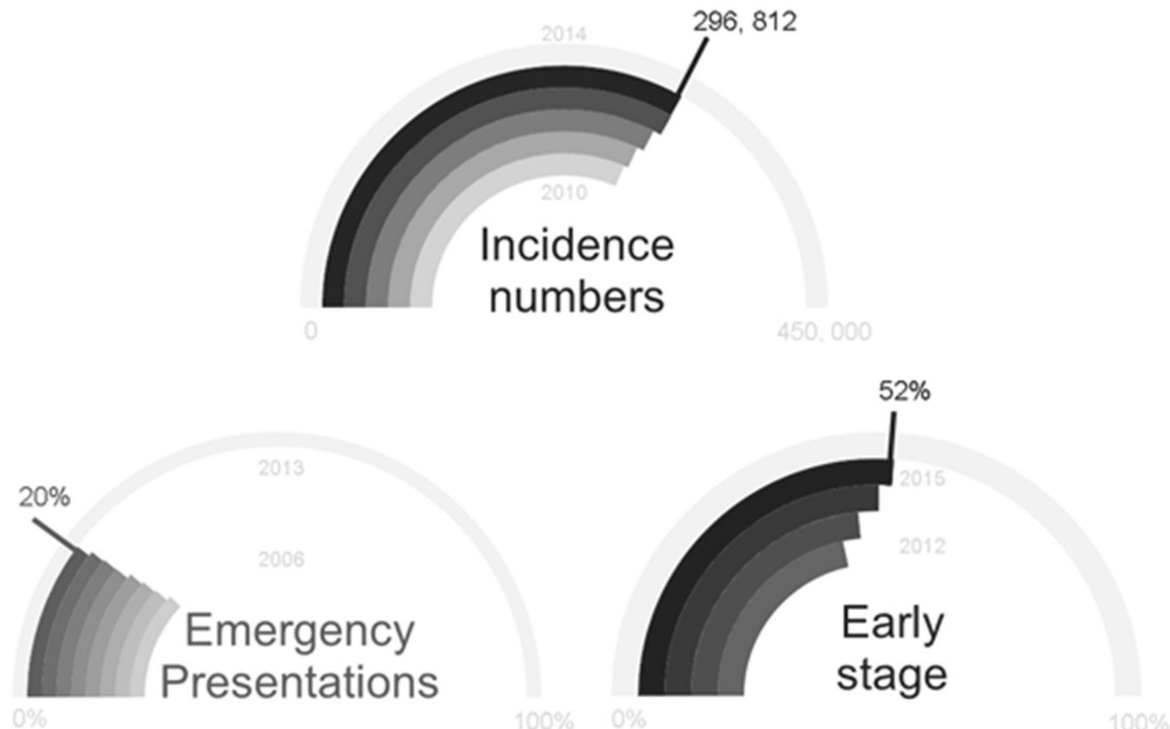
The numbers are getting better



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There's more evidence: take emergency admissions

- Emergency admission percentages have fallen from 25% in 2006 to below 20% in 2022
- That's ~16,000 emergencies avoided p.a.
- The emergency admissions improvements alone amount to 1,000s of deaths averted.

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Sounds easy; think of a route and off you go…

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That’s not how it happened at all!

- All my slides are completely out of sequence

- Most of the cancer identification studies were done before the ‘defining the problem’ work
- I didn’t rush to the Department of Health
- The Department of Health recognized there was a BIG problem, and I was in the right place at the right time

It was a matter of redrawing the route time after time… thinking of impact from the start

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So, I can’t say ‘you can make a BIG splash with your work if you follow what I did,’ but…

If you’re in a hole, stop digging… unless it’s a gold mine!

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